Slide 7

Now I will explain the System Architecture and Data Modeling of the Taraqqub platform.

The first diagram shows the workflow of the system. It starts when a citizen submits a report about an infrastructure issue, such as flooding or road damage. The report includes information like the location, description, and photo. Then, the AI analyzes the report, verifies it, and determines its priority level. After that, the report is stored in the database and automatically sent to the appropriate authority. Finally, the authority reviews the report, takes action, and updates the status so the citizen can track the progress.

The second diagram shows the system architecture. Our platform follows a three-tier architecture. The first layer is the Frontend, which is the interface used by citizens and authorities. The second layer is the Backend, which processes requests, manages system logic, and connects all services together. The third layer is the Database, where all reports, users, feedback, and weather information are stored securely.

The third diagram is the ER Diagram, which shows how the data is connected inside the database. The main entity is the Report, which contains details about the reported issue. Each report is linked to a User who created it. We also have Feedback, where users can share their opinions about the platform, and Tips, which provide safety advice based on weather conditions and location.

Overall, this design allows the platform to manage reports efficiently, support AI analysis, improve communication between citizens and authorities, and provide faster responses during weather-related emergencies.

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slide 8

In this slide, I will explain the Operational Workflow and Data Flow of the Taraqqub platform.

This diagram shows how information moves through the system, starting from the citizen and ending with the authority response.

The process begins when a citizen notices an infrastructure issue, such as flooding, a damaged road, or a blocked drainage system. The user opens the Taraqqub platform and submits a report. The report includes important information such as a description of the issue, the GPS location, and a photo of the damage.

At the same time, the platform collects real-time weather data from the OpenWeatherMap API. This weather information helps the system understand the environmental conditions related to the reported issue.

After the report is submitted, it is sent to the AI Analysis Engine. This is one of the main innovations in our platform. The AI analyzes both the image and the text provided by the user. It checks whether the report is valid, identifies the type of issue, and determines its severity level. The AI can also filter spam or irrelevant reports, reducing the workload for authorities.

Once the report has been verified, it moves to the Smart Routing Engine. This component automatically determines which government authority should receive the report. The routing decision is based on factors such as the location of the issue and its category. For example, road damage may be sent to the municipality, while another type of issue may be directed to a different authority.

After routing, the verified report is displayed on the Authority Dashboard. Through this dashboard, authorities can view reports, monitor priorities, and take the necessary actions. The dashboard helps them manage incidents more efficiently and respond faster during emergencies.

All report information is stored securely in the MongoDB database. The database keeps records of users, reports, weather information, feedback, and report status updates. This ensures that data is organized and can be accessed whenever needed.

The final step is the feedback loop. After the authority reviews the report and takes action, the report status is updated in the system. The citizen can then track the progress of the report and see whether it is pending, under review, or resolved. This improves transparency and builds trust between citizens and authorities.

slide 10

In this slide, I will present the main results of our project.

The survey showed that many users face communication problems when reporting infrastructure issues, which can delay responses.

The chart also shows that most people get weather updates from social media, highlighting the need for a centralized platform.

In addition, 68% of participants strongly supported having one platform for reporting infrastructure issues, and 90% trusted reports more when they included photos and location information.

The evaluation results showed high scores in ease of use, usability, and intention to use, with all scores above 4 out of 5.

Overall, these findings show that Taraqqub improves communication, increases reporting efficiency, and helps authorities respond more effectively during weather emergencies.

slide 11

In conclusion, Taraqqub successfully provides a smart and unified platform for reporting infrastructure issues caused by weather conditions in Oman.

The platform combines real-time weather monitoring, citizen reporting, AI-powered validation, and smart routing into one integrated system. This helps improve communication between citizens and authorities and supports faster decision-making during emergencies.

The survey and evaluation results showed strong user acceptance, high usability, and a positive intention to use the platform. Users found the system easy to use and believed it could improve the reporting process and emergency response.

In addition, the platform increases transparency by allowing users to track the status of their reports and receive updates. This helps build trust between the public and government authorities.

Overall, Taraqqub contributes to public safety, improves disaster management, and supports Oman's digital transformation goals. It demonstrates how modern technologies such as AI and real-time data integration can be used to create practical solutions for real-world challenges.

Slide 12

In this slide, I will discuss the future enhancements of Taraqqub.

One of our future plans is to develop a mobile application. This will allow users to submit reports and receive notifications more easily from their smartphones.

We also aim to integrate the platform with government and emergency management systems. This will improve communication and help authorities respond more efficiently during emergencies.

Another enhancement is the integration of drones and IoT devices. These technologies can provide real-time environmental data and support faster monitoring of affected areas.

In addition, we plan to develop custom-trained AI models to improve report verification, issue classification, and prediction accuracy.

Finally, we aim to support multiple languages to make the platform accessible to a wider range of users across Oman.

Overall, these future enhancements will improve the platform's accessibility, intelligence, scalability, and overall impact, helping Taraqqub become a more powerful solution for public safety and disaster management.